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$$\frac{V_A}{n_A} = \frac{V_B}{n_B}$$

$$n_A = \left(\frac{n_A}{V_A}\right) V_B$$

$$= \frac{1.5\ \text{mol}}{37.6\ \cancel{dm^3}} \times 16.5\ \cancel{dm^3}$$

$$= 0.66\ \text{mol}$$

49. n = 1.82 mole

$V = 5.43\ dm^3$

$T = 69.5 + 273 = 342.5$

P = ?

$PV = nRT$

$$P = \frac{nRT}{V}$$

$$1.82\ \text{mol} \times$$

$$0.821\ \cancel{dm^3}\ \text{atm mol}^{-1}\ \text{K}^{-1}$$

$$P = \frac{\times\ 342.5\ \text{K}}{5.43\ \cancel{dm^3}}$$

$$P = 94.25\ \text{atm}$$

50.

$P_1 = 1.2$ atm

$T_1 = 18^0$ C + 273 = 291 K

$T_2 = 85^0$ C + 273 = 358 K

P_2 = ?

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$P_2 = \left(\frac{P_1}{T_1}\right) \times T_2$$

$$= \frac{1.2\ \text{atm}}{291\ \cancel{K}} \times 358\ \cancel{K}$$

$$P_2 = 1.48\ \text{atm}$$

51. $T_1 = 8\ ^0C + 273 = 281$ K

$P_1 = 6.4$ atm $\quad V_1 = 2.1$ ml

$T_2 = 25^0$ C + 273 = 298 K

$P_2 = 1$ atm $\quad V_2$ = ?

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2}$$

$$= \frac{6.4\ \cancel{atm} \times 2.1\ \text{ml} \times 298\ \cancel{K}}{281\ \cancel{K} \times 1\ \cancel{atm}}$$

$$V_2 = 14.25\ \text{mol}$$

52. கொடுக்கப்பட்டவை,

$V = 154.4 \times 10^{-3}\ dm^3$,

P = 742 mm of Hg

T = 298 K $\quad$ m = ?

$$n = \frac{PV}{RT} = \frac{742\ \text{mm Hg} \times 154.4 \times 10^{-3}\ \text{L}}{62\ \text{mm Hg L K}^{-1}\text{mol}^{-1} \times 298\ \text{K}}$$

$$= 0.006\ \text{mol}$$

$$n = \frac{\text{நிறை}}{\text{மோலார் நிறை}}$$

நிறை = n × மோலார் நிறை

$= 0.006 \times 2.016$

$= 0.0121\ g = 12.1\ mg$

53.

$$\frac{\gamma_{\text{வாயு}}}{\gamma_{N_2}} = \frac{t_{N_2}}{t_{\text{வாயு}}} = \sqrt{\frac{m_{N_2}}{m_{\text{வாயு}}}}$$

$$\frac{84\ \text{sec}}{192\ \text{sec}} = \sqrt{\frac{14\ \text{g mol}^{-1}}{m_{\text{வாயு}}}}$$

$$= \frac{14\ \text{g mol}^{-1}}{m_{\text{வாயு}}}$$

$$m_{\text{வாயு}} = 14\ \text{g mol}^{-1} \times \left(\frac{192\ \text{sec}}{84\ \text{sec}}\right)^2$$

$$m_{\text{வாயு}} = 73.14\ \text{g mol}^{-1}$$